Constructive Scienter: An Animal-Law Answer to the AI Responsibility Gap

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ABSTRACT

This article recovers an overlooked common law doctrine, the scienter action's heightened liability for the keeper of a dangerous animal, and brings it to bear on the opaque algorithmic systems through which the state now makes consequential decisions about individuals. It argues that the less a public body can explain a coercive decision, the more it must justify it. Where public power, algorithmic opacity, and an affected person's right to reasons converge, the system's opacity can neither discharge that duty nor excuse its evasion. Under the scienter action, knowledge of a creature's propensity strengthens rather than severs the keeper's responsibility. Carried into public law, that logic yields a new attributional concept, 'constructive scienter', the deployer's knowledge of what it cannot know. So framed, the much-discussed 'responsibility gap' is, in public law, no gap at all. The dominant response of AI legal personhood misreads the problem, conflating the exclusion that personhood may remedy in animal law with the evasion it would enable in AI governance. The argument runs through *State v Loomis*, with implications for administrative decision-making and judicial review.

KEYWORDS algorithmic accountability; automated decision-making; animal law; scienter; administrative law

1. Introduction

The state owes a reason to those over whom it exercises power. Yet that obligation is treated as oddly easy to suspend once the decision runs through a machine: a public body detains a person and, asked why, can answer only that no one knows, the decision having issued from a system it cannot itself read. That inability is usually received as a regrettable limit on accountability. This article argues that the less a public body can explain its coercive decision, the more it must justify it. Specifically, where public power, algorithmic opacity, and a right to reasons converge, the state's duty of justification is heightened, and the system's opacity can neither discharge that duty nor excuse its evasion. The duty travels with the decision to deploy, not with the technology deployed: opacity the state has knowingly assumed is a feature of its own conduct, and it tightens the very obligation it is most often invoked to escape.

The common law has faced this difficulty before, in an unlikely place. Long before machine learning, the scienter action confronted it in the form of a dangerous animal: a nonhuman agent whose partial independence threatened to diffuse human responsibility. Its response was to hold the keeper answerable in proportion to what he knew of the creature's propensities, so that knowledge of the danger deepened the keeper's responsibility.[1] The modern form of that difficulty has a familiar name, the 'responsibility gap': a consequential decision issues from a system no human fully predicts, harm follows, and no one, it is said, can be held to account.[2] The deeper worry is that public power increasingly secures its ends through technological design, in ways that slip the demand for justification on which the rule of law depends.[3] What travels from it is less about animals than its attributional logic, which

[1] For the scienter action, see Glanville Williams, *Liability for Animals* (CUP 1939) 267–300. The principle is codified in the Animals Act 1971, s 2(2), and affirmed in *Mirvahedy v Henley* [2003] UKHL 16, [2003] 2 AC 491, where the House of Lords calibrated liability to the keeper's awareness of the animal's propensities. The doctrine is set out in full in section 3.

[2] See Andreas Matthias, 'The Responsibility Gap: Ascribing Responsibility for the Actions of Learning Automata' (2004) 6 Ethics and Information Technology 175. For the criminal-law variant, see John Danaher, 'Robots, Law and the Retribution Gap' (2016) 18(4) Ethics and Information Technology 299; and for a taxonomy, Filippo Santoni de Sio and Giulio Mecacci, 'Four Responsibility Gaps with Artificial Intelligence' (2021) 34 Philosophy and Technology 1057. For the responsibility gap within a public-law and human-rights frame, see Karen Yeung, *A Study of the Implications of Advanced Digital Technologies (Including AI Systems) for the Concept of Responsibility Within a Human Rights Framework* (Council of Europe 2019).

[3] The diagnosis is Roger Brownsword's: 'Technological Management and the Rule of Law' (2016) 8(1) Law, Innovation and Technology 100; Roger Brownsword, *Law 3.0: Rules, Regulation and Technology* (Routledge 2020); and, on the displacement of normative regulation by technological management, Roger Brownsword, 'In the Year 2061: From Law to Technological Management' (2015) 7(1) Law, Innovation and Technology and *Law, Technology and Society: Re-imagining the Regulatory Environment* (Routledge 2019). On the rule-of-law and fundamental-rights stakes of public-sector AI, see also Nathalie A Smuha, 'Beyond the Individual:

burdens the keeper in proportion to what he knew, without making the animal a legal person. From that move the article draws 'constructive scienter', the deployer's knowledge of what it cannot know, and shows that, so understood, the responsibility gap is in public law no gap at all. The gap it closes is one of justification: opacity leaves intact the state's obligation to account for a coercive decision, whatever remains unsettled about compensating the harm such systems do. The personhood debate and the animal-law doctrine enter in supporting roles. The first is a framing the argument must set aside; the second, the doctrine that supplies the answer.

When algorithmic systems produce troubling outcomes, legal critique often centres on bias and opacity.[4] One prominent response would recognise sophisticated artificial systems as legal persons. Scholars have debated the conditions for AI to hold rights,[5] traced the patterns of AI personhood,[6] and explored how a disaggregated bundle of rights might fill accountability gaps.[7] In public law, however, turning to personhood distracts more than it illuminates. Legal personality can function as a tool for 'liability management' that shields institutional actors from the consequences of their own choices.[8] The pressing question is not who can be held liable. It is whether the state may rely on algorithmic outputs while evading its own non-delegable duties of reasoned decision-making.[9]

Governing AI's Societal Harm' (2021) 10(3) Internet Policy Review; Mireille Hildebrandt, *Smart Technologies and the End(s) of Law: Novel Entanglements of Law and Technology* (Edward Elgar 2015)

[4] Kate Crawford and Jason Schultz, 'AI Systems as State Actors' (2019) 119 Columbia Law Review 1941; Cary Coglianese and David Lehr, 'Regulating by Robot: Administrative Decision Making in the Machine-Learning Era' (2017) 105 Georgetown Law Journal 1147; Solon Barocas and Andrew D Selbst, 'Big Data's Disparate Impact' (2016) 104(3) California Law Review 671, 674–75.

[5] Lawrence B Solum, 'Legal Personhood for Artificial Intelligences' (1992) 70(4) North Carolina Law Review 1231.

[6] Claudio Novelli and others, 'AI as Legal Persons: Past, Patterns, and Prospects' (2025) 52 Journal of Law and Society 533, 535.

[7] Visa AJ Kurki, *A Theory of Legal Personhood* (OUP 2019) 13–14.

[8] Joanna J Bryson and others, 'Of, For, and By the People: The Legal Lacuna of Synthetic Persons' (2017) 25(3) Artificial Intelligence and Law 273, 286.

[9] The non-delegable duty of care is the closest common-law analogue to the attributional move the scienter action makes. In both contexts, the law holds that certain responsibilities cannot be shed by interposing an

Faced with nonhuman beings capable of independent action, the common law developed doctrines, most notably the scienter action, that burdened the human relation of use according to the keeper's knowledge of the animal's propensity to harm.[10] Scienter's responsibility-burdening logic has drawn comparatively little notice in the AI literature. Recent animal/AI scholarship has moved in a different direction. Visa Kurki and Jeff Sebo identify a structural parallel between animal and AI personhood debates and ask whether personhood might be extended to AI.[11] Meanwhile, other scholars have shown the person/property divide to be inadequate in both fields, prompting intermediate legal frameworks such as 'quasi-property'[12] and 'living property'[13] under which animals remain owned but the human relation of use is governed by heightened duties. Some AI scholars have suggested that AI may similarly require intermediate status.[14] Yet whether the response takes the form of intermediate status or full personhood, the underlying problem remains: personhood does not do the same work in the two fields. In animal law, personhood is invoked to remedy *exclusion*: the law is asked to recognise the intrinsic interests of sentient beings unjustly submerged in the property category. In AI governance, personhood would construct a legal subject that absorbs the accountability that ought to remain with the deploying institution, thereby facilitating *evasion*. A move that makes sense in one setting would aggravate the problem in the other.[15]

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intermediary—whether a contractor or an algorithm—between the duty-holder and the person to whom the duty is owed. See *Woodland v Essex County Council* [2013] UKSC 66.

[10] Halsbury's Laws of England (4th edn, vol 2(1), Butterworths 1991) 620. See also Philip H Osborne, *The Law of Torts* (4th edn, Irwin Law 2011) 356–58.

[11] Kurki (n 6); Jeff Sebo, *The Moral Circle: Who Matters, What Matters, and Why* (WW Norton 2025).

[12] Angela Fernandez, 'Not Quite Property, Not Quite Person: A "Quasi" Approach for Nonhuman Animals' (2019) 5(1) Canadian Journal of Comparative and Contemporary Law 155.

[13] David Favre, 'Living Property: A New Status for Animals Within the Legal System' (2010) 93 Marquette Law Review 1021.

[14] See, eg, Joffrey Baeyaert, 'Beyond Personhood' (2025) Technology and Regulation 355, 360, proposing a hybrid status of limited, context-specific legal recognition for AI.

[15] I take it that present-generation algorithmic systems are not sentient, and I do not claim they share moral equivalence with animals. Nor do I claim that animal law provides a seamless blueprint: intermediate categories in animal law often entrench human domination rather than end it. Yet for current public-law uses of AI, this imperfect burdening of the human beneficiary is doctrinally superior to AI personhood. When personhood

This article makes three contributions, each to a different debate. To the literature on the responsibility gap, it shows that in public law there is no gap to fill: the common law's attributional logic already places responsibility on the institution that knowingly deploys an opaque system. To the debate over legal personhood, it offers a portable distinction between the *exclusion* personhood may rightly remedy and the *evasion* it must not be allowed to enable. And to administrative law, it supplies constructive scienter, the deployer's knowledge of what it cannot know, and from it a burdened-use test for when an opaque public deployment heightens the state's duty to give reasons.[16]

Each of these claims converges on a practical setting: public-sector procurement of proprietary AI. When a state body contracts for an algorithmic tool whose internal logic is shielded by trade secrecy, the procurement decision becomes the moment at which opacity is knowingly introduced into the exercise of public power.[17] From that point forward, the rule-of-law demand for explainability grows more urgent because the affected individual's capacity to challenge a decision depends on understanding its basis.[18] The state cannot shelter behind vendor secrecy: when a supplier's intellectual property claims defeat a citizen's right to reasons, the state has delegated a constitutional function to a commercial entity with no public-law obligations. Certain public-law obligations (due process, the duty to give reasons, the obligation not to act irrationally) cannot be shed by interposing a technical intermediary between the decision-maker and the person affected. The scienter logic this article develops

creates liability sinkholes, animal law's intermediate forms do the opposite: they intensify the obligations of the human deployer in direct proportion to the nonhuman entity's capacity to cause harm.

16 Jessica Eisen, 'Animals in the Constitutional State' (2018) 15(4) International Journal of Constitutional Law 909, 911–14. For the contrasting critique, see Frank Pasquale, 'A Rule of Persons, Not Machines: The Limits of Legal Automation' (2019) 87(1) George Washington Law Review 1, 4.

17 Deirdre K Mulligan and Kenneth A Bamberger, 'Procurement as Policy: Administrative Process for Machine Learning' (2019) 34 Berkeley Technology Law Journal 778. Mulligan and Bamberger demonstrate that procurement is the critical governance intervention point for algorithmic systems, because it is where institutional choices about interpretability, auditability, and risk tolerance are made—or defaulted.

18 *Padfield v Minister of Agriculture, Fisheries and Food* [1968] AC 997 (HL); *R (Osborn) v Parole Board* [2013] UKSC 61. Both decisions affirm that the rule of law requires public bodies to furnish intelligible reasons when their decisions affect individual rights, a requirement that algorithmic mediation does not displace.

offers an account of why those duties travel with the decision to deploy and not with the technology deployed.[19]

The argument proceeds in three parts. Section 2 shows why the person/property binary constrains the governance of emergent nonhuman systems, how personhood serves different functions in animal law and AI governance, and why, as *State v Loomis* shows, proprietary algorithms already generate the accountability gaps that AI personhood would widen. Section 3 develops the scienter action and constructive scienter to argue that the unpredictability of nonhuman entities should strengthen the public deployer's obligations. Section 4 extends the analysis beyond criminal sentencing to administrative decision-making and sets out the doctrinal implications.

2. Personhood and its limits

2.1. Singularist theory and bundle theory

In both animal law and AI governance, personhood marks the boundary between objecthood and legal significance. A 'thing' enters the law through the rights and interests of others, while a 'person' may enter as a subject in its own right. But that division rests on a prior question that Ngaire Naffine identifies as foundational to legal ordering: 'Who is law for?'[20] To classify an entity as a person or thing is to decide whether it appears in law as an end in itself

[19] The argument complements recent scholarship on systems theory and located accountabilities, which similarly resists the temptation to isolate fault in the technical artefact. See e.g. Henry L Fraser and Nicolas P Suzor, 'Locating Fault for AI Harms: A Systems Theory of Foreseeability, Reasonable Care and Causal Responsibility in the AI Value Chain' (2025) 17(1) Law, Innovation and Technology 103. Fraser and Suzor argue that fault inquiries should proceed from the understanding that AI harms arise from socio-technical conditions rather than discrete technical errors, and that attributions of responsibility should focus on actors with knowledge and control over systemic risk factors.
[20] Ngaire Naffine, *Law's Meaning of Life: Philosophy, Religion, Darwin and the Legal Person* (Hart Publishing 2009) 1.

or as a means to the ends of human users. The question becomes harder to answer when legal personhood is treated as a single status that either attaches in full or not at all.

Visa Kurki calls the orthodox view 'singularist': an entity either has the full complement of legal capacities (standing, ownership, the capacity to sue and be sued) or it has none.[21] Singularism operates as an 'inference ticket':[22] once an entity is labelled a person, the law tends to import a familiar set of legal incidents without asking which of them the entity can meaningfully bear.[23] In the AI context, this is especially troublesome. It allows legal subjectivity to be attached to a system for functional reasons without first asking whether the entity has interests that make such recognition intelligible. This omission matters because the prior question, whether AI could ever be an interest-bearing being, remains unresolved. Although Geoffrey Hinton has suggested that present-generation AI may already possess sentience,[24] the dominant scholarly view remains that present-generation AI systems do not suffer injury or possess wills.[25] Yet a singularist approach could still attach to them legal capacities designed to protect those kinds of interests, insulating human users from liability for AI activity.

Kurki's 'bundle theory' offers a more pragmatic alternative. On this view, legal personhood is a cluster of legal incidents, including claim-rights, privileges, powers, and immunities together with their Hohfeldian correlatives, that can be distributed selectively depending on context.[26] If personhood is decomposable, AI systems need not be forced into a

[21] Kurki (n 6) 6, 12; see also Novelli and others (n 5) 536.
[22] Kurki (n 6) 31.
[23] Kurki (n 6) 25.
[24] See '"Godfather of AI" Geoffrey Hinton: The 60 Minutes Interview' (YouTube, 8 October 2023), in which Hinton states that such systems have experiences 'in the same sense as people do'.
[25] Sentience remains the most important benchmark in this context because it bears most directly on the capacity to suffer harm or possess interests. Although Hinton's position remains a minority view, it illustrates how quickly the sentience debate is evolving.
[26] Kurki (n 6) 25, 31. See also Wesley Newcomb Hohfeld, 'Some Fundamental Legal Conceptions as Applied in Judicial Reasoning' (1913) 23(1) Yale Law Journal 16. Novelli and others arrive at a similar conclusion through their distinction between 'singularist' and 'clustered' conceptions: Novelli and others (n 5) 536–38.

choice between full legal subjectivity and mere property. The bundle approach can allocate duties and liabilities directly to the human institutions that deploy and benefit from such systems, without first resolving what AI is.

Animal law shows this is already a settled practice. Animals may be protected by statutory welfare duties and other legal obligations without possessing personhood.[27] As Saskia Stucki notes, animals may hold simple rights without holding the more expansive rights associated with full legal subjectivity.[28] The law already recognises intermediate positions that the person/thing divide cannot describe.[29] Duties grow heavier because the entity under human control is neither a full legal subject able to protect its own interests nor a mere object whose effects can be disowned. The law need not invent AI personhood to impose heightened obligations of justification on the institution that chooses to rely on a nonhuman system whose outputs carry coercive force.

2.2. Personhood's divergent functions

Personhood has historically been a flexible doctrinal tool deployed for pragmatic governance rather than moral worth.[30] While the common law could in principle extend personhood to AI for functional reasons, conflating its animal-law and AI-law deployments obscures a crucial difference in the problem each domain is trying to solve.

In animal law, the push for personhood responds to exclusion. The premise is that a sentient being, possessing intrinsic interests and a capacity for suffering, has been unjustly submerged within the category of property. For example, when the Nonhuman Rights Project

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[27] *Nonhuman Rights Project, Inc v Breheny*, 38 NY (3d) 555 (2022) 15–16.
[28] Saskia Stucki, 'Towards a Theory of Legal Animal Rights: Simple and Fundamental Rights' (2020) 40 Oxford Journal of Legal Studies 533, 552–56.
[29] Kurki (n 6) 14; see also Fernandez (n 10) 255; Favre (n 11) 1021.
[30] Katherine B Forrest, 'The Ethics and Challenges of Legal Personhood for AI' (2024) 133 Yale Law Journal Forum 1175, 1177; Joffrey Baeyaert, 'Beyond Personhood: The Evolution of Legal Personhood and Its Implications for AI Recognition' (2025) Technology and Regulation 356, 361.

brought habeas corpus petitions on behalf of Happy the elephant in 2018, the aim was to compel courts to ask whether confinement itself could constitute a legal wrong to the animal.[31] The New York Court of Appeals rejected that claim on the ground that personhood presupposes the capacity to bear social duties as well as rights,[32] but the dissent argued that the logical progression of common law runs from extending habeas corpus to women, children, and enslaved persons, all once denied the writ because they were deemed chattel, to granting an elephant the same right.[33] The dissent's logic is recognitional: it proceeds by analogy with prior expansions of personhood to categories of beings whose subjective interests the law once refused to see. That form of reasoning has little purchase on a risk-assessment algorithm.

The drive towards AI personhood addresses an entirely different problem. Few commentators have suggested that a predictive risk-assessment tool like COMPAS suffers a cognisable wrong when it is retrained or shut down.[34] The push for AI personhood is an attempt to manufacture a liable entity when an autonomous system causes harm and human actors appear unaccountable, much as the corporation was devised to carry liabilities that would otherwise rest on the individuals behind it.[35] As Joanna Bryson warns, legal personality presents state and corporate actors with 'liability management' opportunities, allowing those who deploy AI systems to shield themselves from the consequences of their conduct behind an artificial person.[36] In public law, the problem is less that the state lacks a

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[31] Steven M Wise, *Rattling the Cage: Toward Legal Rights for Animals* (Perseus 2000), 2–5. See also Gary L Francione, *Animals, Property, and the Law (*Temple University Press 1995).
[32] Breheny (n 21) 4–9.
[33] ibid 4–9, 70 (Wilson J, dissenting). The majority characterised the comparison as 'odious': ibid 10.
[34] COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is a proprietary recidivism risk-assessment tool developed by Northpointe (now Equivant) and used in several US jurisdictions to inform bail, sentencing, and parole decisions. Its use in criminal sentencing is examined below: see *State v Loomis*, 881 NW 2d 749 (Wis 2016).
[35] Forrest (n 29) 1196–98.
[36] Bryson and others (n 7) 274–75, 286. That dynamic doomed the EU Parliament's 2017 proposal for 'electronic personhood'. See Novelli and others (n 5) 549.

liable entity than that it diffuses its own responsibility through technical processes.[37] Granting AI personhood would deepen this crisis, manufacturing a fictitious legal subject that absorbs the accountability the state is required to bear. The distinction can be put as a test of general application. Whenever personhood, or some lesser legal subjectivity, is proposed for a nonhuman entity, the prior question is which of two things it would do: remedy an exclusion, by recognising interests the law has wrongly refused to see, or license an evasion, by manufacturing a subject onto which a responsible human actor can offload accountability. Personhood earns its place in the first case and forfeits it in the second. The test travels wherever the proposal does, from animals to rivers to artificial systems.[38]

One might object that AI personhood remains attractive in light of forward-looking arguments about moral agency and consciousness—and there is little doubt that some future AI systems might warrant recognition on these grounds.[39] Yet the present crisis involves predictive AI models already deployed by courts and agencies that are devoid of subjective experience and are implausible candidates for moral patiency. To defer reform until the ontology of machine consciousness is resolved is to grant the state an indefinite reprieve from its constitutional duties of due process and explanation. Public law needs doctrinal instruments calibrated to the pathology of algorithmic opacity in state action. Animal law, as section 3 argues, already contains one.

[37] Pasquale (n 14) 4–8.

[38] See, eg, the Te Awa Tupua (Whanganui River Claims Settlement) Act 2017 (NZ) s 14, the first statute to confer legal personality on a river.

[39] For instance, Jeff Sebo's argument for an expanded 'moral circle', including potentially infinite AI systems, cautions against the historical error of drawing the boundaries that decide who matters too narrowly. See Sebo (n 9). This view reflects precisely the conflation of how and why personhood is deployed in the animal and AI contexts. However, it remains contingent on the capacity of future-generation AI for sentience. Similarly, Katherine Forrest's suggestion that AI may develop an 'emotive quality' raises genuine questions about synthetic consciousness. See Forrest (n 24) 1177, 1185. More recently, Gellers and Holy-Luczaj have argued that intelligent machines can and should be morally considerable, drawing on environmental ethics to challenge the exclusion of artefacts from the moral circle. See Joshua C Gellers and Magdalena Holy-Luczaj, 'Consider the Xenobot: Moral Status for Intelligent Machines Revisited' (2025) Law, Innovation and Technology 17.2: 444-66.

2.3. Why AI personhood is the wrong response

Once an algorithmic system is imagined as a distinct legal subject, it becomes easy to describe harmful outcomes as flowing from the system rather than from the human choice to deploy it. But the problem does not originate there. Recent jurisprudence shows that algorithms' current status as property already enables a comparable evasion.[40] *State v Loomis* is the clearest illustration. The Wisconsin Supreme Court upheld the use of the proprietary COMPAS risk-assessment tool in criminal sentencing.[41] Eric Loomis argued that his due process rights were violated because the algorithm's proprietary nature prevented him from assessing how it produced his high-risk recidivism score,[42] and although the court mandated warning labels, it relocated the burden of algorithmic opacity onto the defendant.[43] Loomis was expected to bear the consequences of an automated output he could neither inspect nor rebut, an inversion of the principle that the state must justify its exercises of coercive power. *Loomis* thus exposes a broader pattern: the state outsources constitutional functions to private vendors and then relies on the algorithm's status as intellectual property to defeat due process claims.[44] Wisconsin extracted the predictive administrative utility of COMPAS while avoiding fuller due process explanation. If mere propertyhood already enables evasion, granting the AI legal subjectivity would deepen the problem, transforming a barrier to

[40] Jenna Burrell, 'How the Machine "Thinks": Understanding Opacity in Machine Learning Algorithms' (2016) 3(1) Big Data and Society 1, 3–5.
[41] *State v Loomis*, 2016 WI 68, 371 Wis 2d 235, 881 NW 2d 749. See also Katherine Freeman, 'Algorithmic Injustice: How the Wisconsin Supreme Court Failed to Protect Due Process Rights in State v Loomis' (2016) 18 North Carolina Journal of Law and Technology 75.
[42] *Loomis* (n 34) para 5. ProPublica's investigation suggests that Black defendants were more likely to be falsely labelled high risk while white defendants were more likely to be falsely labelled low risk: Jeff Larson and others, 'How We Analyzed the COMPAS Recidivism Algorithm' (ProPublica, 23 May 2016).
[43] *Loomis* (n 34) paras 13–15, 51. The expert witness Dr David Thompson testified that sentencing courts have little information about how COMPAS analyses risk and that presenting its outputs without this context 'mis-inform[s] the court'.
[44] Sonia K Katyal, 'Private Accountability in the Age of Artificial Intelligence' (2019) 66(1) UCLA Law Review 54, 60–65.

transparency (trade secrecy) into a recognised juridical boundary (an independent legal person).[45]

Some scholars propose intermediate paths to bypass the person/property binary. Joffrey Baeyaert, for example, describes a constrained 'e-person' status for high-stakes contexts like autonomous vehicles. In this model, an AI system would function as a limited 'legal envelope', able to hold assets or assume contractual obligations so as to facilitate compensation without displacing human responsibility. The proposal is explicitly fiduciary: developers would stand in a trustee-like relation to the system.[46] Baeyaert is attentive to the danger that legal personality might become a vehicle for blame-shifting, and he treats the status as strictly provisional with human fallback structures.[47]

Even so, the model retains a major vulnerability. Once any form of legal subjectivity is introduced, however qualified, a new juridical intermediary stands between the deploying institution and the person injured by its use. The fiduciary overlay mitigates that risk, but it does not remove it.[48] In a case like *Loomis*, designating COMPAS as an e-person would still make it easier for responsibility to be filtered through the system itself. The state could argue that the relevant duties attach in the first instance to the entity through which the decision was mediated. The vendor could identify that entity as the formal site of liability. And the defendant would confront an additional legal layer before reaching the institution whose choice to rely on the tool produced the harm. The difficulty is that once legal subjectivity enters the picture, even in carefully cabined form, it creates a structure that institutions may

[45] That the barrier is a vendor's trade secret only sharpens the point. As Rebecca Wexler has argued of the criminal process more broadly, permitting a developer's intellectual property to defeat a person's access to the basis of a decision subordinates a constitutional interest to a commercial one, and does so without the justification such a privilege would have to earn anywhere else. Rebecca Wexler, 'Life, Liberty, and Trade Secrets: Intellectual Property in the Criminal Justice System' (2018) 70 Stanford Law Review 1343.
[46] Baeyaert (n 24) 379.
[47] ibid 359, 381.
[48] ibid 355, 379–80.

later use to diffuse accountability instead of bearing it directly. The history of corporate personhood gives reason to take that possibility seriously: legal fictions introduced for narrow and practical ends rarely remain confined to them.[49]

3. The scienter action and constructive scienter

If AI personhood either misdescribes the problem or makes it worse, the next question is whether the common law already offers another way to think about nonhuman intermediaries whose conduct matters without diffusing human responsibility. The scienter action shows that it does. In a closely related setting, the common law confronted the basic problem algorithmic governance now presents: how to prevent a nonhuman intermediary's relative independence from becoming a means of escaping human responsibility.

3.1. The scienter action and the autonomous residuum

Under the common law, owning a living animal is not the same as owning an inanimate object. A hammer strikes only when swung; a dog may bite of its own volition. The living entity under human ownership possesses a capacity for independent, legally consequential action that inert property lacks.[50] Early English law classified animals as chattels, but a distinct body of rules emerged to govern the *autonomous residuum* of animal behaviour: the irreducible margin of independent action no amount of training could eliminate.[51] When a farmer's ox gored a neighbour, courts did not solve the problem by declaring the animal a legal person capable of being sued. Instead, the law placed heavier obligations on the human

[49] Forrest (n 24) 1196–98.
[50] The French Civil Code was amended in 2015 to carve animals out of the pronouncement that all property was either movable or immovable. New art 515-14 defines animals as '*êtres vivants doués de sensibilité'* (living beings gifted with sentience). While animals remain classified under the regime of property, they are now recognised as more than mere moveables. Code civil, art 515-14, as amended by Loi n° 2015-177 du 16 février 2015.
[51] Osborne (n 8) 355.

owner *because of* the nonhuman entity's capacity to act on its own. The property right was burdened from within by the nature of the thing owned: the more autonomous the entity, the heavier the duties attached to keeping it.

Case law grounded this principle in concrete liability. In *Filburn v People's Palace and Aquarium Co*, the Court of Appeal held the keeper of a performing elephant strictly liable when the animal struck and injured a boy passing its stall.[52] The court held that an elephant belonged to the class *ferae naturae*, whose capacity for uncontrolled harm the law imputed categorically; keeping the creature at all sufficed for strict responsibility. For domesticated animals, liability attached only upon actual or constructive notice of the individual animal's propensity to cause the type of harm suffered. The Animals Act 1971 codified this distinction, preserving the scienter principle under section 2(2).[53] In *Mirvahedy v Henley*, the House of Lords reaffirmed that the statutory scheme calibrated liability to the keeper's awareness of the animal's propensities.[54] The Law Commission had earlier recommended preserving the scienter principle in statutory form on the ground that it served a distinct attributional function, connecting the keeper's knowledge to the keeper's responsibility in a way that ordinary negligence could not.[55]

[52] *Filburn v People's Palace and Aquarium Co* (1890) 25 QBD 258 (CA). The boy's injuries resulted from the elephant's trunk striking him as he passed its stall. The court did not inquire into the animal's prior behaviour: classification as *ferae naturae* was sufficient.

[53] Animals Act 1971, s 2(2). The subsection preserves the common law scienter by imposing strict liability where damage is of a kind the animal was likely to cause (or which was likely to be severe), the likelihood was attributable to characteristics not normally found in that species (or found only at particular times or circumstances), and the keeper knew or ought to have known of those characteristics. See also Glanville Williams, *Liability for Animals* (CUP 1939) 267–300.

[54] *Mirvahedy v Henley* [2003] UKHL 16, [2003] 2 AC 491 [10] (Lord Nicholls). Lord Nicholls emphasised that the statutory scheme 'achieves a reasonable balance' between the interests of keepers and the public precisely because it calibrates the stringency of liability to the keeper's awareness of the animal's propensities.

[55] Law Commission, Civil Liability for Animals (Law Com No 13, 1967) paras 14-18. The Commission recommended preserving the scienter principle in statutory form precisely because it served a distinct attributional function: connecting the keeper's knowledge to the keeper's responsibility in a way that ordinary negligence could not.

The scienter action crystallised that intuition. At common law, animals were divided between those *ferae naturae* (wild by nature, such as wolves, whose dangerousness was imputed by category) and those *mansuetae naturae* (domesticated, such as dogs or cattle, presumed harmless until the keeper came to know otherwise).[56] For wild animals, the keeper's liability was absolute: the decision to harbour an inherently unpredictable creature was sufficient ground for strict liability regardless of the precautions taken. For domesticated animals, strict liability attached once the keeper had scienter, meaning actual or constructive knowledge of the specific animal's tendency to cause damage of the kind suffered.[57] Knowledge of the propensity was the very basis of the keeper's heightened responsibility. As Glanville Williams demonstrated in his comprehensive account, the keeper's liability was never predicated on fault in the modern sense but on the voluntary assumption of a known risk, a principle whose force does not depend on the biological nature of the entity kept.[58]

What makes the scienter instructive is less its black-letter rule than the direction of its causal logic. In ordinary tort, an intervening autonomous act (say, an animal's unprovoked lunge) typically breaks the chain of proximate causation and relieves the original actor of liability. Scienter inverts that default: the animal's autonomy does not interrupt the chain but strengthens it. The more dangerous and unpredictable the animal, the stricter the liability imposed on the human who benefits from keeping it.[59] Peter Cane's distinction between outcome-responsibility and role-responsibility clarifies what is at work: scienter liability is not primarily about the keeper's causal contribution to the specific harm but about the

[56] ibid.
[57] ibid.
[58] Glanville Williams, *Liability for Animals* (CUP 1939) 267-300, 312-15. Williams traced the scienter action's development from medieval English law and demonstrated that the keeper's liability was never predicated on fault in the modern sense but on the voluntary assumption of a known risk, a principle that survives in the Animals Act 1971. The wild-animal category is itself imperfect: as Williams observed, some animals *ferae naturae*, such as rabbits and pigeons, are harmless and carry no presumption of danger. See Law Commission, Civil Liability for Animals (Law Com No 13, 1967) para 15.
[59] ibid 359.

keeper's institutional position as the party who voluntarily assumed the risk of keeping a partially independent source of danger.[60] The common law treated unpredictability as the trigger because a person who unilaterally introduces a source of autonomous risk into the community for their own benefit cannot externalise the costs of that risk onto those who had no say in the matter.[61] In the terms of the present debate, this is an answer to the responsibility gap given in advance: when the literature on autonomous systems treats unpredictability as what opens the gap, the scienter action treats unpredictability that has been knowingly assumed as what closes it. The intuition is an old one in the theory of strict liability. Namely, a party who imposes a non-reciprocal risk on others for its own benefit must answer for it. The point does not depend on the biological nature of the thing kept.[62]

3.2. From scienter to constructive scienter

A further question is latent in the doctrinal taxonomy: where do contemporary AI systems sit within the wild/domestic division? While any neat mapping of AI systems onto this binary oversimplifies the matter, the better classification is *mansuetae naturae*. A state agency procuring a machine-learning model for sentencing typically knows the system's intended purpose, the task to which it has been confined, and the general contours of its operational domain. Such a system is closer, doctrinally, to the trained domestic animal whose general course of conduct is foreseeable while its particular behaviour is not, than to the wild predator whose dangerousness the common law imputed by category alone. The classification matters

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[60] Peter Cane, *Responsibility in Law and Morality* (Hart Publishing 2002) 144-51. Cane distinguishes between outcome-responsibility and role-responsibility, a distinction that illuminates why the deployer's institutional position matters more than the algorithm's causal contribution to the harm.
[61] Samir Chopra and Laurence F White, *A Legal Theory for Autonomous Artificial Agents* (University of Michigan Press 2011) 130–31. Chopra and White are among the few to have connected the scienter action to AI governance, and the most sustained of them, though their treatment of the doctrine is brief. For foundational commentary, see Halsbury's Laws (n 8) 620; Osborne (n 8) 356–58.
[62] The proposition that strict liability answers the imposition of non-reciprocal risk is George P Fletcher's: 'Fairness and Utility in Tort Theory (1972) 85 Harvard Law Review 537. Its extension to a knowingly opaque public deployment is the argument advanced here.

because domestic-animal liability turns on scienter (actual or constructive knowledge of the specific propensity to cause damage of the kind suffered) rather than attaching automatically as with wild animals. The doctrinally consequential question is therefore what it means, as a matter of common law attribution, for a state deployer to bear scienter of an opaque system's propensities.

The analogy pressed here should be distinguished from a related move now gathering pace in private law. A growing body of tort scholarship has reached for the wild-animal and abnormally dangerous activity categories to argue that those who build or release autonomous systems should bear strict liability for the harms the systems cause.[63] That literature and the present argument share an intuition, that unpredictability is a reason for stricter responsibility, but they diverge on the question asked and the doctrine used. The tort accounts are concerned with compensation, i.e., with finding a defendant who must pay when an autonomous product injures a stranger. The argument here is concerned with justification: with the public body's constitutional duty to give the person it binds reasons it can stand behind. And where the tort accounts use scienter in its ordinary sense, the keeper's knowledge of a specific dangerous trait, the argument here turns on a different and more demanding idea: the deployer's knowledge of its own incapacity to know. It is to that idea that the analysis now turns.

The knowledge element is satisfied in constructive rather than specific form. A public institution that adopts a model whose internal reasoning is opaque even to its designers knows, at the moment of deployment, that the system will generate consequential outputs it

[63] For the wild-animal and abnormally dangerous activity framings of AI liability in private law, see, eg, Trent Kannegieter, 'Nondeterministic Torts: A Technical Approach to AI Liability' (forthcoming) 135 Yale Law Journal; Mihailis E Diamantis, 'Vicarious Liability for AI' (2023) 99 Indiana Law Journal 317; and Renee Henson, 'Applying Strict Liability to Artificial Intelligence as an Abnormally Dangerous Activity' (2024) 96 Temple Law Review 349. These accounts ask who must compensate the victim of an autonomous system, whereas the present argument asks how the state must justify a coercive decision, and turns on the deployer's knowledge of its own incapacity to know rather than knowledge of a specific dangerous propensity. The most sustained prior attempt to connect animal-keeper liability to AI, by Chopra and White, likewise remains within private law and does not develop the constructive form of the knowledge element.

cannot anticipate in the individual case. This general awareness of opacity is itself constructive scienter of a dangerous propensity, foreseeable from the outset, to produce determinations the deployer cannot justify on the particular facts before it. On this account, opacity does not defeat scienter but supplies its content. The deployer's knowledge of what it cannot know is the constructive form of the doctrine.[64] Where even this attenuated form of scienter cannot be sustained (for instance, where the deployer could not articulate what comparable performance for the system would look like, or what would count as a deviation from baseline) the *mansuetae* framework breaks down, and the system reverts to the *ferae naturae* default, categorical dangerousness the deployer has invited by choosing to keep an instrument it cannot meaningfully monitor.[65]

Knowledge of what one cannot know is, arguably, not knowledge at all. Yet the objection fails, because the law has long imputed knowledge to a party who arranges not to have it. Under the doctrine of wilful blindness, the common law fixes a person with knowledge of a fact they suspected and deliberately declined to confirm, reasoning that calculated ignorance is no less culpable than knowledge itself.[66] Constructive scienter is the near relation of that doctrine, and in one respect its graver case. The wilfully blind defendant could have looked and chose not to. The public deployer of an opaque system has gone further: he has knowingly taken on an instrument whose particular determinations no one can

[64] Chopra and White (n 47) 130.
[65] Chopra and White (n 45) 127-33. Their observation that a custodian who becomes aware of a particular AI agent's dangerous propensities should attract heightened obligations anticipates the constructive-scienter argument developed here, though they did not develop the concept of constructive knowledge or extend it to public-law deployments.
[66] The doctrine is old and cross-cutting. In the criminal law, wilful blindness is treated as knowledge: *Westminster City Council v Croyalgrange Ltd* [1986] 1 WLR 674 (HL). In commercial law, 'blind-eye knowledge' requires a suspicion together with a deliberate decision not to confirm it: *Manifest Shipping Co Ltd v Uni-Polaris Insurance Co Ltd* (The Star Sea) [2001] UKHL 1, [2003] 1 AC 469 [112]-[116] (Lord Scott). In the United States, see *Global-Tech Appliances Inc v SEB SA*, 563 US 754 (2011), requiring a subjective belief in a high probability of the fact and a deliberate effort to avoid confirming it. For a corporate-liability treatment of AI knowledge proceeding along different lines, see Mihailis E Diamantis, 'Employed Algorithms: A Labor Model of Corporate Liability for AI' (2023) 72 Duke Law Journal 797.

confirm, and bound individuals by them all the same. If the law imputes knowledge to the party who would not look, it must hold answerable the party who has arranged that no one can. The deployer’s culpable knowledge is not of the output it cannot foresee but of the condition it has chosen: that it will issue coercive determinations it will be unable to justify. So understood, the 'constructive' here is not borrowed from the law of notice, where a party is fixed with knowledge it ought to have acquired. The deployer's knowledge is actual: knowledge of a structural incapacity he has chosen.

Applied to *Loomis*, this reframing exposes the Wisconsin court's error in treating COMPAS's opacity as a background condition rather than as a feature of the state's own conduct. Wisconsin adopted a proprietary risk-assessment tool whose internal weighting was opaque even to those deploying it, in a setting where its outputs would directly shape sentencing. On the *mansuetae* framing developed above, the state's knowledge of that opacity is itself constructive scienter of the system's foreseeable propensity to produce determinations the state could not defend on the individual facts before it. Just as the keeper of a known-dangerous dog cannot say 'I did not know that this bite would happen on this day, at this particular moment, to this particular person', the state that knowingly deploys an opaque predictive model cannot say 'we cannot explain what the algorithm produced, at this time, affecting this person', nor that a vendor's trade secret excuses the silence. The inability to explain is the foreseeable consequence of the decision to deploy, and it tightens rather than severs the duty of justification the state owes the individual before it.[67]

One might further object that the scienter parallel misidentifies the responsible party, since COMPAS was built by Northpointe rather than by the state. Yet the duty of a scienter defendant has never run with formal title to the animal. It runs with whoever keeps and

[67] Loomis (n 34) para 23.

benefits from the nonhuman entity's use and holds the practical capacity to control its exposure to others. The innkeeper who harboured a guest's dog, and the zoo operator who displayed a borrowed animal, each attracted scienter liability, because each stood in the relevant functional relation to the nonhuman source of risk. Wisconsin procured COMPAS, wove its outputs into sentencing, and drew coercive value from its predictions. Northpointe's proprietary ownership was legally incidental, because the state occupied the keeper's role toward the defendant. On that framing, the trade secret claims *Loomis* treated as a meaningful constraint on due process review can no longer perform the shielding work the court allowed them to do. Opacity must travel forward into the reasons the state offers, rather than backward into the private arrangements that produced it.

A contrasting disposition from another jurisdiction confirms the point. In the Dutch *SyRI* case, the District Court of The Hague struck down the System Risk Indication algorithm, used by the Dutch government to detect welfare fraud, under Article 8 of the European Convention on Human Rights.[68] The court held that the state bore the burden of demonstrating that the opacity of its algorithmic profiling system was proportionate to the legitimate aim pursued, and that the government had failed to discharge that burden. Unlike the Wisconsin Supreme Court in *Loomis*, the Hague court treated algorithmic opacity as a feature of the state's own conduct that heightened its justificatory obligations. That is the move the scienter logic demands. The contrast is the organising thread of the cases gathered here. Placed side by side, *Loomis* and *SyRI* mark the two paths a court may take when opacity meets public power: Wisconsin received the opacity as a background condition to be worked around, and let the duty of justification slacken, whereas the Hague court treated the same

[68] Rechtbank Den Haag (District Court of The Hague) 5 February 2020, ECLI:NL:RBDHA:2020:1878 (*SyRI*). The court held that the System Risk Indication algorithm violated art 8(2) ECHR because its opacity prevented effective scrutiny of the risk profiles it generated, and the government had failed to demonstrate that the interference was proportionate.

opacity as something the state had itself introduced, and held the duty taut. Robodebt and the visa streaming tool fall on the Wisconsin side of that line. This article argues that the Hague side is the one constructive scienter requires.[69]

4. Burdened use: the duty in public law

4.1. Beyond sentencing: how the logic travels

The argument developed through *Loomis* is not confined to criminal sentencing. Its force derives from a general proposition about state action: wherever a public body deploys an opaque algorithmic system in a setting where legal justification is owed to the affected person, the deployer's knowledge of opacity engages the same attributional structure. The duty to give reasons does not vary with the subject-matter of the decision: it attaches wherever the state exercises power that alters an individual's legal position.[70] Only the intensity of review changes. Two further settings carry the burdened-use logic beyond criminal sentencing, each with a doctrinal specificity that *Loomis* alone cannot supply.

The two settings that follow confirm a structure already established. The first is automated welfare administration. Australia's Robodebt scheme, which operated between 2015 and 2019, used an automated income-averaging method to generate debt notices to social security recipients.[71] The Royal Commission found that the Department of Social Services and Services Australia deployed the automated method knowing that it lacked a

[69] The *SyRI* decision has been widely discussed. See, eg, Janneke Gerards and Raphaèle Xenidis, 'Algorithmic Discrimination in Europe: Challenges and Opportunities for EU Equality Law' in European Network of Legal Experts in Gender Equality and Non-Discrimination (European Commission 2021) 28–30.
[70] *R (Osborn) v Parole Board* [2013] UKSC 61 [68] (Lord Reed). Lord Reed affirmed that procedural fairness requires that an individual affected by a decision be given the opportunity to make representations, and that the decision-maker provide reasons that engage with the substance of those representations. The principle applies with particular force where the decision engages the liberty of the subject.
[71] *Report of the Royal Commission into the Robodebt Scheme* (Commonwealth of Australia 2023). The income-averaging method was conceded to be unlawful, and debts raised solely on that basis declared invalid, in *Prygodicz v Commonwealth of Australia (No 2)* [2021] FCA 634 (Murphy J).

lawful basis, and that senior officials systematically failed to seek legal advice that might have confirmed the deficiency.[72] The Commission concluded that officials relied on the system's technical complexity to deflect ministerial and judicial scrutiny.[73] On the constructive-scienter analysis, the department's knowledge that its automated method produced legally unreliable outputs was itself constructive scienter of a propensity to cause unjustified harm. That knowledge did not excuse the deployment, but constituted the ground on which the duty to justify each notice should have been most stringent. What Robodebt demonstrates, beyond *Loomis*, is the institutional dimension of constructive scienter: it was a deliberate administrative choice to deploy an automated process whose legal frailty was known to those who authorised it. The burdened-use framework captures this as a heightening of the deployer's duty, not a diffusion of blame across the value chain.

The second is immigration decision-making. From 2015 to 2020, the UK Home Office operated a 'visa streaming algorithm' that classified visa applicants by nationality into risk categories (red, amber, or green) determining the level of scrutiny each application received. The criteria governing classification were never disclosed to applicants. In 2020, the Joint Council for the Welfare of Immigrants brought a legal challenge alleging that the tool discriminated on the basis of nationality in breach of the public sector equality duty under section 149 of the Equality Act 2010.[74] The Home Office abandoned the tool before the claim was adjudicated. The visa algorithm is instructive here for the structural feature it shares with COMPAS, whatever the merits of the discrimination allegation: the Home Office could not explain to any individual applicant why their application had been streamed into a

[72] Procurement is the juncture at which interpretability and auditability are settled or defaulted: Deirdre K Mulligan and Kenneth A Bamberger, 'Procurement as Policy: Administrative Process for Machine Learning' (2019) 34 Berkeley Technology Law Journal 773, 778-82.
[73] *Robodebt Royal Commission Report* (n 72).
[74] R (Joint Council for the Welfare of Immigrants) v Secretary of State for the Home Department, Pre-Action Protocol Letter (26 June 2020). The Home Office abandoned the Streaming Tool after JCWI's challenge alleged that sorting applicants by nationality into risk categories constituted unlawful discrimination under the Equality Act 2010, s 149 (public sector equality duty).

particular category, because the criteria were opaque even within the department. On the constructive-scienter account, the Home Office's awareness that its streaming tool operated on undisclosed and internally unjustifiable criteria is itself knowledge of a propensity to produce determinations that cannot be defended in individual terms. The burdened-use framework does not require proof of discriminatory intent. It requires only that the deployer knew, at the moment of adoption, that it was introducing a system whose outputs it could not justify to those affected, and that it deployed the system nonetheless.[75]

These examples confirm that the accountability gap *Loomis* exposed is structural rather than domain-specific. The common feature is the conjunction of three elements: public power, opacity, and an affected person's entitlement to reasons. Wherever those three elements converge, the burdened-use logic applies. The state's knowledge that it has introduced an opaque intermediary into the exercise of power that requires justification is sufficient to engage the heightened duty.

4.2. Doctrinal implications: what burdened use requires

If public law were to adopt the attributional logic the scienter action supplies, several doctrinal consequences would follow. Each represents a recalibration of existing common-law and public-law principles in the direction the scienter analogy points. This section sets out, with some concreteness, what constructive scienter would demand of a public body that deploys an opaque algorithmic system in a setting where justification is owed.

First, the state cannot rely on proprietary opacity as an excuse for deficient reasons. Where a public body deploys an algorithmic tool whose internal operations are shielded by

[75] The episode is documented in detail by Jack Maxwell and Joe Tomlinson, whose study of the Home Office's automation experiments shows the streaming tool to be one instance of a pattern in which opacity and the absence of individualised reasons travelled together. Jack Maxwell and Joe Tomlinson, *Experiments in Automating Immigration Systems* (Bristol University Press 2022).

trade secrecy, the inability to explain the tool's output to the affected individual is a deficiency in the state's justificatory performance. The common law's increasing recognition that a duty to give reasons is an incident of lawful decision-making supports this position.[76] Rebecca Williams has made the strongest version of this case, arguing that administrative law's existing toolkit (e.g. the duty to give reasons, the requirement of relevance, the prohibition on fettering discretion) is already capable of addressing algorithmic decision-making without new statutory frameworks, provided courts are willing to apply those principles to the deployer's choice of tool as well as to the decision itself.[77] Williams is right that the tools exist. But her account leaves open the question of *why* the deployer's choice of an opaque tool should attract the level of scrutiny she proposes. The conventional answer is good administrative practice, but that framing treats opacity as a contingent deficiency rather constitutive of the deployer's conduct. Constructive scienter supplies the missing attributional logic, grounding the heightened duty of justification in the deployer's prior knowledge that the system will produce outputs it cannot justify to the affected individual. While Williams provides the doctrinal toolkit, constructive scienter explains why the toolkit must be applied with the force she advocates. On the Wednesbury standard, a decision-maker who cannot explain how a relevant consideration was taken into account has, by definition, failed to demonstrate that the decision was not reached unreasonably.[78]

[76] cf Mark Elliott, 'Has the Common Law Duty to Give Reasons Come of Age Yet?' [2011] Public Law 56, 59–63. Elliott argues that the common law increasingly treats the duty to give reasons as an incident of lawful decision-making rather than an exceptional remedy, a trajectory that supports the position that opacity should be treated as a deficiency in the state's justificatory performance rather than a background condition.
[77] Rebecca Williams, 'Rethinking Administrative Law for Algorithmic Decision Making' (2022) 42(2) Oxford Journal of Legal Studies 468, 478-85.
[78] *Associated Provincial Picture Houses Ltd v Wednesbury Corporation* [1948] 1 KB 223 (CA); *Council of Civil Service Unions v Minister for the Civil Service* [1985] AC 374 (HL) (*CCSU*); *R (Keyu) v Secretary of State for Foreign and Commonwealth Affairs* [2015] UKSC 69. The standard of review in English administrative law varies with context, but the common thread is that the decision-maker must be able to offer intelligible reasons for its decision. Where the decision engages fundamental rights, the standard of scrutiny intensifies toward proportionality.

Second, procurement choices become legally significant. On the burdened-use account, the moment at which a public body contracts for a proprietary system it knows to be opaque is the moment at which constructive scienter attaches. If the state later cannot justify the system's outputs in individual cases, the deficiency traces back to a choice the state itself made: the choice to acquire an instrument whose reasoning it could not inspect. That choice cannot be treated as exogenous to the lawfulness of the eventual decision. Procurement is the point at which opacity is knowingly introduced, and the burdened-use logic holds that knowledge to be constitutive of the heightened duty, not merely relevant to it. In judicial review terms, the decision to procure an opaque system for a high-stakes function is itself a reviewable exercise of public power, because it determines the state's capacity to discharge its justificatory obligations in every subsequent case.

Third, if a system's outputs cannot be justified in relation to the individual case, the defect lies not only in the evidence available to the decision-maker but in the state's prior choice to rely on a tool that forecloses individual justification. The conventional framing treats unjustifiable outputs as an evidentiary problem (the decision-maker lacked sufficient information) but the burdened-use framework recharacterises it as a structural one. The state chose to decide by means that make individual justification impossible, and that choice is itself reviewable. This recharacterisation aligns with Reuben Binns's argument that public justification requires reasons accessible to those affected, a requirement that algorithmic opacity structurally undermines.[79] This is, in substance, the lesson of the common law's turn toward a culture of justification, under which the legitimacy of a decision depends on its being justified to the person it binds and not merely justifiable in the abstract.[80] It also

[79] Reuben Binns, 'Algorithmic Accountability and Public Reason' (2018) 31(4) Philosophy and Technology 543, 546-50.

[80] See e.g. *Canada (Minister of Citizenship and Immigration) v Vavilov* 2019 SCC 65, [2019] 4 SCR 653 (the 'culture of justification'): the exercise of public power must be justified to the individuals subject to it, and not merely justifiable in the abstract.

answers the difficulty, pressed by Jennifer Cobbe, that a review trained on the process by which a decision was reached struggles to reach a decision an opaque system has produced: the burdened-use framework relocates the reviewable moment to the choice to deploy.[81]

Fourth, and most consequentially: in the highest-stakes settings the inability to explain does not merely weaken the state's justification but defeats the lawfulness of the deployment itself. For instance, where the stakes for the individual are highest (criminal sentencing, deprivation of liberty, removal of subsistence benefits) the proposition that the state may deploy a tool it cannot explain faces a direct conflict with the rule-of-law demand that coercive power be exercised for reasons the subject can engage with.[82] In those settings, burdened use implies that the state must either secure sufficient interpretability to discharge its justificatory duty or refrain from relying on the system's output. The scienter logic suggests that the weight of the duty is proportionate to the deployer's knowledge of the risk and the severity of the consequences for the affected person, a calibration that mirrors the proportionality analysis already familiar to public law.

The cumulative force of these implications can be stated as a single test, one the preceding sections have been building toward. Call it the burdened-use test. A claimant invokes it by establishing four conditions: (1) that the decision was an exercise of public power over them; (2) that it was mediated by a system the deployer knew, at the moment of procurement, to be opaque; (3) that the deployer thereby had constructive scienter of the system's propensity to produce determinations it could not justify in the individual case; and (4) that reasons were owed. On proof of these, the burden shifts to the deployer to justify the particular output, or to show that the deployment was attended by safeguards (meaningful

[81] Jennifer Cobbe, 'Administrative Law and the Machines of Government: Judicial Review of Automated Public-Sector Decision-Making' (2019) 39 Legal Studies 636.
[82] Danielle Keats Citron, 'Technological Due Process' (2008) 85 Washington University Law Review 1249, 1256-63.

human oversight, interpretability, a genuine avenue of challenge) adequate to discharge the heightened duty. Where not even a baseline of expected performance can be articulated, the framework reverts to the categorical default the common law reserved for the inherently dangerous.

Consider how a claim grounded in constructive scienter would proceed in judicial review. A claimant affected by an opaque algorithmic decision (say, a benefits recipient whose entitlement was reassessed by an automated income-averaging tool, as in Robodebt) would challenge it on conventional public-law grounds, citing irrationality, procedural unfairness, or proportionality where fundamental rights are engaged. The constructive-scienter framework recharacterises what the claimant must show and what the decision-maker must answer. The claimant would argue that the decision-maker's prior awareness of the system's opacity (evidenced by procurement records, impact assessments, or their absence) was itself constructive knowledge of a propensity to produce unjustifiable outcomes. That knowledge engaged a heightened duty of justification at the point of deployment. The decision-maker's later inability to explain the specific output to the claimant is therefore not an evidentiary shortfall but evidence that the duty was breached in the decision to deploy. The burden then shifts to the decision-maker to show either that the output was justifiable in the individual case, or that the deployment carried sufficient safeguards (human oversight, interpretability measures, meaningful avenues of challenge) to discharge the heightened duty. The framework introduces no new ground of review. It works through the existing ones, and the scienter analogy's attributional logic strengthens them.

Finally, constructive scienter clarifies the relationship between common-law duties and regulatory compliance. The EU AI Act, which entered into force in 2024, classifies AI systems by risk level and imposes transparency and oversight obligations on deployers of high-risk systems.[83] The Act operates ex ante through prescribed compliance requirements. The scienter logic operates ex post through the attribution of responsibility when an opaque deployment causes harm. A deployer who satisfies the AI Act's requirements for its risk class may still produce unjustifiable outputs in particular cases. The constructive-scienter framework holds that regulatory compliance does not exhaust the deployer's common-law duty of justification. The state's obligation to explain its exercises of coercive power to the affected individual is a constitutional requirement, and it persists regardless of whether the system deployed has been classified as compliant under the Act.[84]

4.3. The limits of the analogy

The objection most likely to press against this argument is that the analogy between animals and algorithms breaks down under the weight of its disanalogies. Whereas animals have embodied agency and natural propensities, algorithms are human-designed decision tools operating within institutions. Scienter developed in response to the hazards of keeping living creatures in an increasingly urbanised society. Algorithms present institutional complexities the scienter action never contemplated, such as distributed design, statistical discrimination, and contested baselines for acceptable performance.

The disanalogies are real. But they do not touch the attributional point. What the article borrows is the attributional move the scienter action makes, unpredictability deepening

[83] Regulation (EU) 2024/1689 of the European Parliament and of the Council of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act) [2024] OJ L 2024/1689, arts 13-14, 86. Article 13 requires high-risk AI systems to be designed for sufficient transparency to enable deployers to interpret and use outputs appropriately. Article 14 mandates human oversight measures proportionate to the risks. Article 86 establishes a right to explanation for individual decisions.

[84] The AI Act's transparency and oversight obligations are complementary to, but distinct from, the common-law argument developed here. The Act operates ex ante through regulatory classification and compliance requirements; the scienter logic operates ex post through the attribution of responsibility when opaque deployments cause harm. The two frameworks converge, however, on a shared premise: that the deployer's knowledge of a system's characteristics should determine the stringency of its obligations. See Margot E Kaminski, 'The Right to Explanation, Explained' (2019) 34 Berkeley Technology Law Journal 189, 197-204.

human responsibility, which is the move public law requires in the face of opaque algorithmic systems. Scienter never turned on the inner life of the animal: it turned on the keeper's knowledge and benefit. In each setting the keeper or deployer benefits from the nonhuman entity's outputs, acts through an instrument whose independence it cannot fully predict, and is tempted to treat that independence as an intervening cause that breaks the chain of responsibility. Scienter shows that the common law resisted that temptation. Opacity in contemporary machine learning is the functional counterpart of the autonomous residuum: the irreducible margin of behaviour the deployer cannot fully predict, knowingly introduced into an environment that affects others' rights. That the margin is statistical rather than muscular does not alter the attributional question.

A different objection cuts the other way. If the work is done by an attributional principle—that one who knowingly introduces an uncontrollable source of risk for private benefit must answer for it—why route the argument through animals at all, when the principle could be stated directly, as the law of non-reciprocal risk and non-delegable duty already does?[85] Because the animal cases are the proof on which the argument rests. They show the common law, squarely faced with an autonomous nonhuman agent whose conduct it could have treated as an intervening cause, declining that escape and fastening the duty on the human who benefited. A bare principle of non-reciprocal risk does not establish that the move is available. The scienter cases do, because they exemplify the law making exactly the move algorithmic accountability now needs, and calibrating it to knowledge in exactly the way constructive scienter requires. The non-delegable duty, the nearest public-law principle, tells us only that a duty cannot be shed, not that its stringency rises with what the duty-holder

[85] The principle that one who imposes a non-reciprocal risk for private benefit must answer for it is George P Fletcher's, 'Fairness and Utility in Tort Theory (1972) 85 Harvard Law Review 537; the principle that some duties cannot be shed by interposing an intermediary is the law of non-delegable duty: *Woodland v Essex County Council* [2013] UKSC 66. Neither, stated as a bare principle, supplies the worked instance the scienter cases provide.

knew. The scienter line alone supplies that graduated calibration. The animal cases are the precedent on which the argument stands.

5. Conclusion

When the state exercises coercive power, its duty to justify that power to the person it binds does not lapse because the decision was routed through a machine. Instead, it grows heavier, and the opacity the state chose to introduce is the measure of how much heavier. That is what this article has called burdened use. The AI personhood debate, at least in public law, begins in the wrong place. It asks what an algorithm *is*, person or property, when the more urgent question is what duty the state owes when it chooses to use one with serious consequences for individual liberty. The distinction between exclusion and evasion shows why a personhood framework developed for one problem cannot simply be carried over to the other, and why animal law points in a more useful doctrinal direction: one that burdens the human relation of use rather than constructing a new legal subject.

To be sure, constructive scienter does not resolve every difficulty algorithmic accountability presents, but the two it bears on are the ones that matter most. The first is a worry about the rule of law: as public power comes to be exercised through technological design rather than through rules, the demand for justification on which it depends can quietly lapse. The second is the concern that algorithmic accountability must be fixed on the actors with effective control over a socio-technical system rather than dispersed across its parts.[86] Constructive scienter answers both: it refuses to let the choice of an opaque instrument become the reason the state need not explain itself, and it fastens the duty on the institution

[86] See Karen Yeung, 'Algorithmic Regulation: A Critical Interrogation' (2018) 12(4) Regulation and Governance 505; and Yeung, *A Study of the Implications of Advanced Digital Technologies (Including AI Systems) for the Concept of Responsibility Within a Human Rights Framework* (Council of Europe 2019).

that made the choice. Much remains to be done: to test how far the burdened-use test travels beyond the settings examined here, to fit it to the EU AI Act's ex ante obligations, to settle the evidentiary thresholds that would make it administrable, and to mark how often its strongest form, that some deployments are unlawful precisely because they cannot be explained, will bind. But the direction is not in doubt. The state's duty to give reasons is not a casualty of automation. It is what automation must be built to serve.

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